INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
SIM1006 - TUTORIAL 4

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1. Show that the given function $\mu(x, y)$ is the integrating factor for the corresponding differential equation. Then, solve the equation.
   (a) $\mu(x, y) = \frac{1}{x^2}$, $\quad x\,dy - y\,dx = 0.$
   (b) $\mu(x, y) = e^x$, $\quad \sin y\;dx + \cos y\;dy = 0.$
   (c) $\mu(x, y) = e^{xy}$, $\quad y^2\,dx + (1 + xy)\,dy = 0.$
   (d) $\mu(x, y) = \frac{1}{x^2+y^2}$, $\quad x\,dx + y\,dy + 4y^3(x^2 + y^2)\,dy = 0.$

2. Solve the following non-exact differential equation.
   (a) $2x\,dx - e^{y-x}\,dy = 0.$
   (b) $x\cosh y\;dy - \sinh y\;dx = 0.$
   (c) $(y + 1)\,dx - (x + 1)\,dy = 0.$
   (d) $(3xe^y + 2y)\,dx + (x^2e^y + x)\,dy = 0$

3. Solve the following homogeneous differential equation.
   (a) $(2x - 3y)\,dx - (2y + 3x)\,dy = 0.$
   (b) $(6x^2 - 7y^2)\,dx - 14xy\,dy = 0.$
   (c) $xy^2\,dy - (x^3 + y^3)\,dx = 0, \quad y(1) = 0.$

4. Solve the following equation with linear coefficient.
   (a) $(x - 2y + 4)dx + (2x - y + 2)dy = 0.$
   (b) $(2x + 3y - 1)dx - 4(x + 1)dy = 0.$
   (c) $(2x + y)dx - (4x + 2y - 1)dy = 0.$
   (d) $(x + y - 3)dx + (x + y + 5)dy = 0, \quad y(1) = 0.$

5. Obtain the value α so that the equation $(xy^2 + \alpha x^2y)dx + (x + y)x^2dy = 0$ is exact and hence, solve the equation.